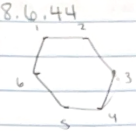
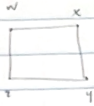
6) 8.6.44





f(1) = w (1,2) ↔ (w,x)

f(2) = x (2,3) ↔ (x,y)

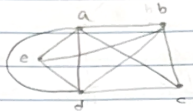
f(3) = y (3,4) ↔ (y,z)

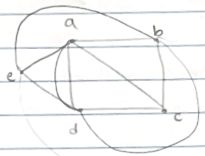
f(4) = z (4,5) ↔ (z,w)

f(5) = w (5,6) ↔ (w,z)

f(6) = z (6,1) ↔ (z,w)

7) 8.7.7





planar ✓

8) By Euler's Formula: |V| − |E| + |F| = 2

6 ↙ (6·4)/2 = 12 ↓

6 − 12 + |F| = 2

|F| = 8 regions

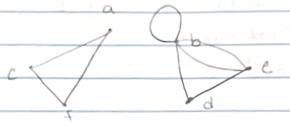
HW9 due 3/8/24

1) 8.5.6



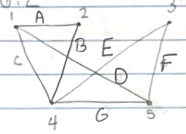
	1	2	3	4	5
1	0	1	1	1	1
2	1	0	1	1	1
3	1	1	0	1	1
4	1	1	1	0	1
5	1	1	1	1	0

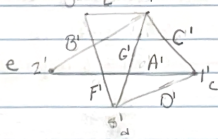
2) 8.5.14



3) 8.5.26 This means that vertex had degree 0, and there are no edges incident on it. This means the graph would have an isolated vertex.

4) 8.6.2





same adjacency matrix:

	1	2	3	4	5
1	0	1	0	1	1
2	1	0	0	1	0
3	0	0	0	1	1
4	1	1	1	0	1
5	1	0	1	1	0

bijection $f: V_1 \to V_2$ $\{a,b,c,d,e\} \xrightarrow{d} \{3,4,1,5,2\}$

bijection g: $\{A',B',C',D',E',F',G'\} \to \{A,B,C,D,E,F,G\}$ multiset of degrees $\{2^2,3^2,4^3\} \to$

5) 8.6.8 G_1 deg G_2 deg

a: 3 e: 2 1: 3 5: 2

b: 3 f: 3 2: 3 6: 3

c: 3 g: 2 3: 4 7: 3

d: 3 h: 3 4: 2 8: 2

$\{2^2, 3^6\} \neq \{2^3, 3^4, 4^1\}$

So G_1 and G_2 are not isomorphic

	1'	2'	3'	4'	5'	$\{2^2,3^2,4^3\}$
1'	0	1	0	1	1	
2'	1	0	0	1	0	
3'	0	0	0	1	1	
4'	1	1	1	0	1	
5'	1	0	1	1	0	